TOPIC 12

Type I and Type II Errors

The two ways a hypothesis test can be wrong, how α and β trade off, and what the power of a test means

WHAT YOU WILL LEARN

- Define Type I and Type II errors and say which is a false positive
- Read the 2 × 2 decision table of truth versus decision
- Link α to the Type I error rate and β to the Type II error rate
- Explain power (1 − β) and the four things that raise it
- Calculate β and power for a z-test, and see the α–β trade-off
- Choose which error matters more in a real situation

NOTE **Where this connects**

Topic 10 (Hypothesis Testing) ended with a decision: reject H_0 or fail to reject H_0. A decision made from a **sample** can be wrong. This topic names the two ways it can go wrong and shows how to control them. The next topic (13, P-value) shows how strongly the data argue against H_0.

1. A decision can be right or wrong

Before the test, the truth is hidden: H_0 is either true or false, and we never find out for certain. After the test we either reject H_0 or fail to reject it. Combining the two gives four possible outcomes, two correct and two wrong.

	H_0 is true (nothing special)	H_0 is false (there is an effect)
We reject H_0	**Type I error** (false positive)	Correct: a real effect is detected (**power**)
We fail to reject H_0	Correct: nothing found, and nothing there	**Type II error** (false negative)

KEY IDEA **Remember it this way**

- **Type I** = reject a **true** null hypothesis. You raise an alarm that should not have sounded (false alarm, false positive).
- **Type II** = fail to reject a **false** null hypothesis. You miss something that is really there (missed detection, false negative).

The class notes say 'accept the null hypothesis' for the second row. Statisticians prefer **fail to reject**, because a sample can never prove H_0 true; it can only fail to provide enough evidence against it (see Topic 10).

2. The class example, translated

The lecture uses a story: you are not sure whether someone likes you back. Make the null hypothesis **H_0: she likes you back**, then see how each error arises.

Truth	Your decision	Outcome
H_0 true: she likes you back	You reject H_0 and do not invite her out	**Type I error**: you rejected a true null and missed your chance
H_0 false: she does not like you back	You accept H_0 and invite her out	**Type II error**: you kept a false null and got an awkward answer
H_0 true	You keep H_0 and invite her	Correct
H_0 false	You reject H_0 and do not invite her	Correct

The story also shows that the two mistakes have different costs. Here the first is more painful (a lost opportunity) and the second is mildly embarrassing. In real problems you must ask which mistake is costlier *in that situation* (Section 8).

3. Type I error and α

- A Type I error is **rejecting a true H_0**: a false positive.
- Its probability is the **significance level α**: P(reject H_0 | H_0 true) = α.
- **You choose α** before the test (often 0.05, 0.01 or 0.10). So the researcher controls how often false alarms occur.
- If α = 0.05 and H_0 is true, then in the long run about 1 test in 20 will wrongly reject it.

A quick simulation

We run 200,000 one-sided z-tests (H_0: μ = 100, σ = 15, n = 36, α = 0.05) on data generated with H_0 **true**. The proportion that reject is **0.0499**, almost exactly α = 0.05. Each of those rejections is a Type I error.

4. Type II error and β

- A Type II error is **failing to reject a false H_0**: a false negative.
- Its probability is **β**: P(fail to reject H_0 | H_0 false).
- Unlike α, β is **not chosen directly**. It depends mainly on the **sample size** and the **variability** (σ) of the data, and also on how big the true effect is and on α.
- A topic that is hard to measure (small or noisy samples) makes Type II errors more likely. That is not the researcher's fault, which is why Type II is usually seen as the smaller worry, but it still costs real discoveries.

DEFINITION **Power of a test**

Power = 1 − β = the probability of **rejecting a false H_0**. This is exactly what a researcher wants. The lecture calls it 'the power of the test'. A common target is power ≥ 0.80.

NOTE **Recall from earlier topics**

- **Standard error:** SE = σ/√n measures how much a sample mean varies from sample to sample (Topic 07).
- **Critical value:** the cut-off for the test statistic (or sample mean) beyond which H_0 is rejected; for a right-tailed test at α = 0.05 it is z = 1.645 (Topics 04 and 10).
- **Effect size:** how far the true value is from the H_0 value (here 105 − 100 = 5). Bigger effects are easier to detect.

5. Seeing α and β together

Take a right-tailed z-test of H_0: $\mu = 100$ against H_1: $\mu > 100$, with $\sigma = 15$ and $n = 36$, so SE = $15/\sqrt{36} = 2.5$. Use $\alpha = 0.05$. The test rejects when the sample mean exceeds the **critical value**.

1. Critical value: $\bar{x}^* = 100 + 1.645 \times 2.5 =$ **104.11**. Reject H_0 if $\bar{x} > 104.11$.
2. Suppose the **truth** is $\mu = 105$. Then $\bar{x} \sim \text{Normal}(105, 2.5)$.
3. $\beta = P(\bar{x} \le 104.11 \mid \mu = 105) = P(Z \le (104.11 - 105)/2.5) = P(Z \le -0.3551) =$ **0.361**.
4. Power = $1 - \beta =$ **0.639**.

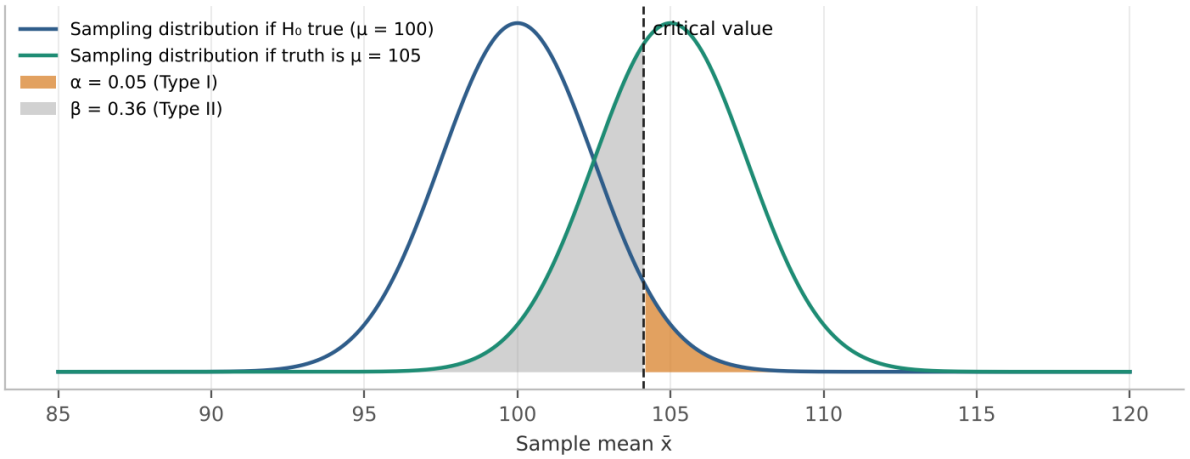


Figure 1. Orange is the Type I error region (α) under H_0. Grey is the Type II error region (β): sample means that fall short of the critical value even though the true mean is 105. The rest of the green curve to the right of the line is the power, 0.64.

Simulation check: generating data with $\mu = 105$ and applying the same test 200,000 times, the proportion rejecting H_0 is **0.639**, matching the calculated power 0.639.

6. The α–β trade-off

For a fixed sample size, **lowering α raises β**. Moving the critical value to the right reduces the orange area but enlarges the grey area. You cannot shrink both errors by shifting the cut-off; you must change the data.

α	Critical value $\bar{x}^*$	β	Power
0.1	103.20	0.236	0.764
0.05	104.11	0.361	0.639
0.01	105.82	0.628	0.372
0.001	107.73	0.862	0.138

Same test as above (true $\mu = 105$). Strict α protects against false alarms but makes real effects harder to detect.

7. How to raise power

Change	Effect on power	Why
Larger sample size n	Increases	SE = σ/√n shrinks, so the two curves overlap less
Less variable data (smaller σ)	Increases	Same reason: smaller SE
Bigger true effect	Increases	The curves are further apart
Larger α	Increases	Wider rejection region, at the price of more Type I errors

In practice the lever you control is **sample size**. The lecture puts it plainly: researchers increase the power of a test by increasing the sample size. The figure shows power for the same z-test (σ = 15, α = 0.05, one-sided) for three true effects.

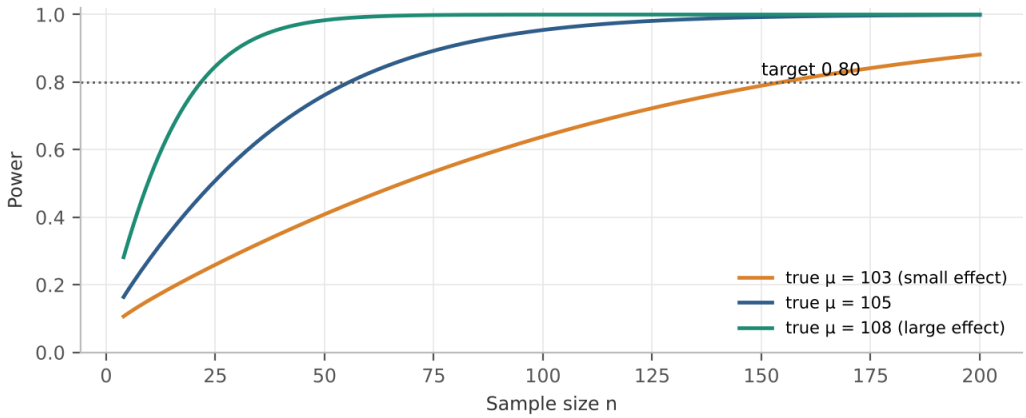


Figure 2. Power climbs with n, and climbs faster when the true effect is bigger.

True mean (H_0: 100)	Sample size for power 0.80
103	155
105	56
108	22

A true mean only 3 points from the null needs about seven times the sample of one 8 points away (155 against 22). Small effects are expensive to detect.

8. Which error is worse?

The lecture says Type I is 'the more serious error'. That is a useful default for scientific claims (do not announce an effect that is not there), but it is **not a law**. The right choice depends on the consequences.

Situation	Type I (false positive)	Type II (false negative)	Which is costlier?
Court trial (H_0: innocent)	Convict an innocent person	Free a guilty person	Type I; the system sets a very high bar
Cancer screening (H_0: healthy)	Healthy person told 'possible cancer'; more tests	Cancer missed	Often Type II; set α larger, accept false alarms
Spam filter (H_0: genuine mail)	Genuine mail sent to spam	Spam reaches inbox	Usually Type I (losing a real email)
Factory quality check (H_0: batch is fine)	Good batch scrapped	Faulty batch shipped	Depends on recall costs
Drug approval (H_0: no benefit)	Useless drug approved	Useful drug rejected	Both serious; trial sizes are planned for power

COMMON MISTAKE **Do not reduce α blindly**

Setting α = 0.001 sounds safe but, with the same data, it can push β above 0.5 so that real effects usually go unnoticed. Decide α **and** the power you need *before* collecting data, then compute the sample size.

9. The same idea in machine learning

Later, when you evaluate classifiers, the same two errors appear under new names: a **false positive** is a Type I error and a **false negative** is a Type II error. Power is called **recall** (sensitivity). The reasoning about costs in Section 8 carries over unchanged.

10. In Python

```
import numpy as np
from scipy import stats

mu0, mu1, sigma, n, alpha = 100, 105, 15, 36, 0.05
se = sigma / np.sqrt(n)
crit = mu0 + stats.norm.ppf(1 - alpha) * se      # reject if xbar > crit
beta = stats.norm.cdf((crit - mu1) / se)
print(round(crit, 2), round(beta, 3), round(1 - beta, 3))

# sample size for power 0.80
d = mu1 - mu0
n_needed = ((stats.norm.ppf(1 - alpha) + stats.norm.ppf(0.80)) * sigma / d) ** 2
print(int(np.ceil(n_needed)))
```

SUMMARY

- Type I error = rejecting a true H_0 (false positive); probability α, chosen by the researcher.
- Type II error = failing to reject a false H_0 (false negative); probability β, driven by n, σ, effect size and α.
- Power = 1 − β = probability of detecting a real effect; aim for about 0.80.
- With n fixed, lowering α raises β. Only more data (or cleaner data) lowers both.
- Increase power mainly by increasing sample size.
- Which error is worse depends on the costs in context.

Practice questions

Try each question before reading the answer underneath it.

Q1. A new drug is tested and a trial wrongly concludes it works when it does not. Which error, and which Greek letter gives its probability?

Answer. Type I error (false positive); α.

Q2. A test has $\beta = 0.25$. What is its power, and what does it mean?

Answer. Power $= 1 - 0.25 = 0.75$. If the effect is real, the test will detect it 75% of the time.

Q3. You halve the significance level from 0.10 to 0.05 and keep everything else the same. What happens to α, β and power?

Answer. α halves; β increases; power decreases.

Q4. Name two ways to reduce β without increasing α.

Answer. Increase the sample size, or reduce the variability of measurements (a better instrument or design).

Q5. In Section 5, how large must n be to reach power 0.80 when the true mean is 105 ($\sigma = 15$, $\alpha = 0.05$, one-sided)?

Answer. $n = ((1.645 + 0.842) \times 15/5)^2 = 55.6$, so $n = 56$.

Q6. For a fire alarm, H_0 is 'there is no fire'. Describe a Type I and a Type II error and say which is more dangerous.

Answer. Type I: alarm rings with no fire (false alarm). Type II: a real fire and no alarm. Type II is far more dangerous, so the alarm is tuned to ring easily (large α).